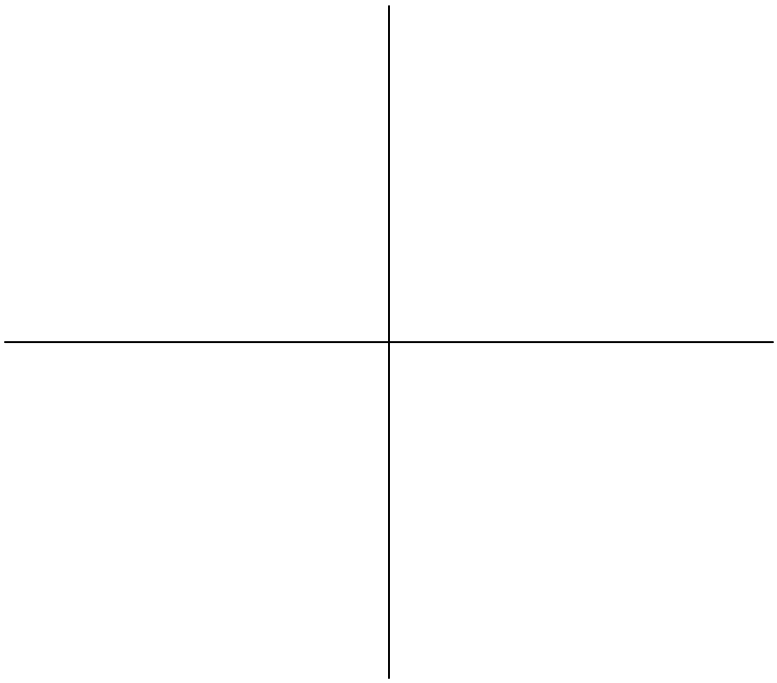
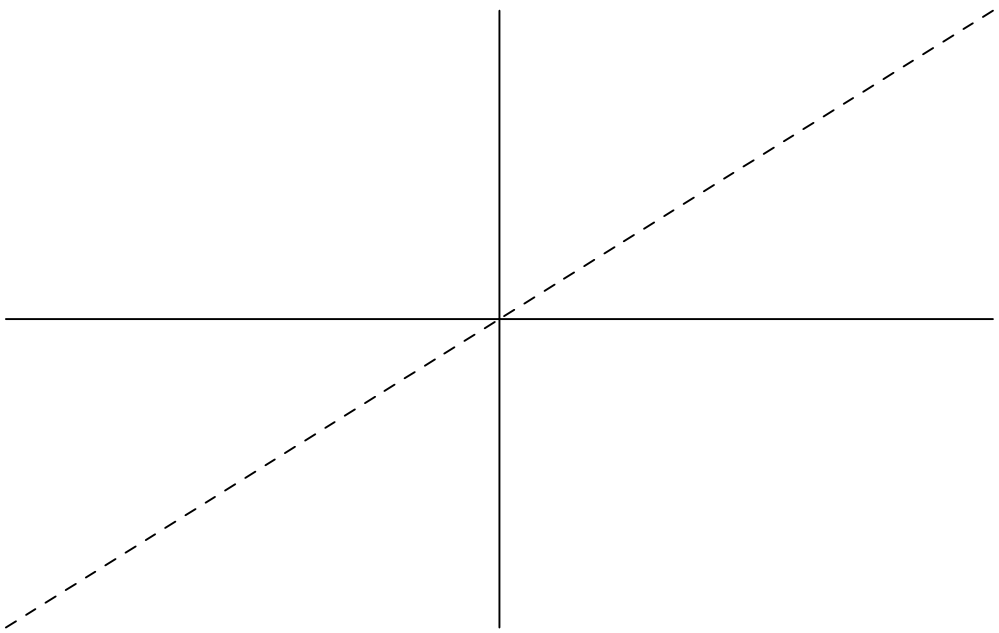
Sec 9.1 3-d coordinate system

Recall how we define points in the 2d system

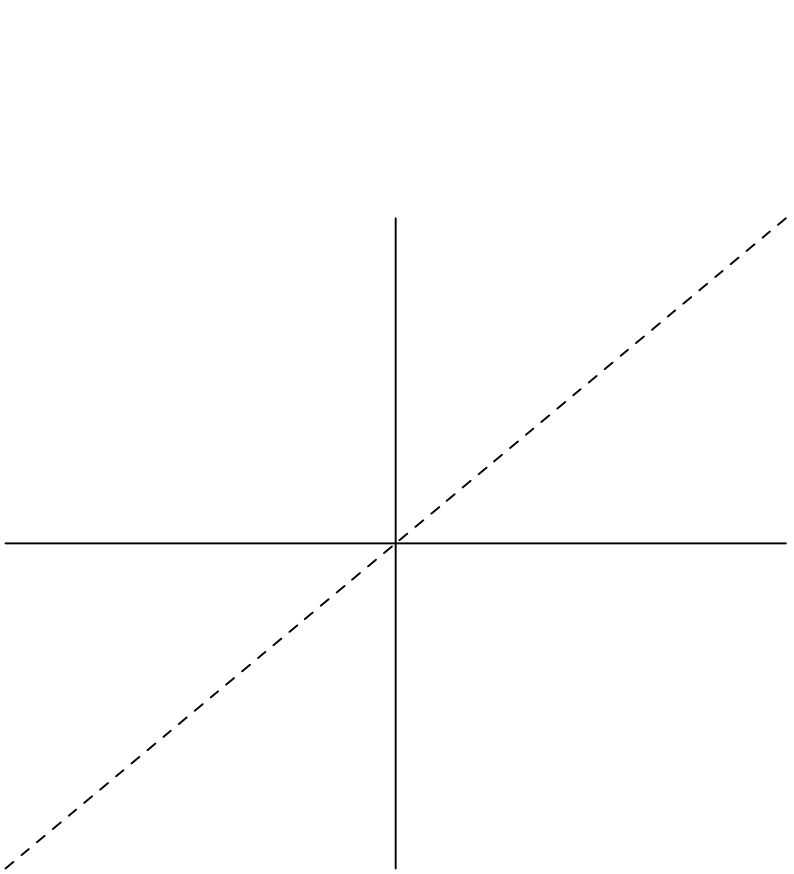


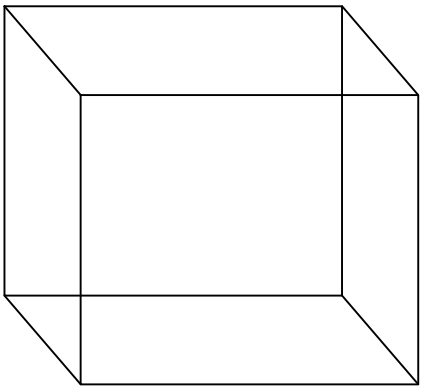
If we want to move to 3-d we just need one more Coordinate!

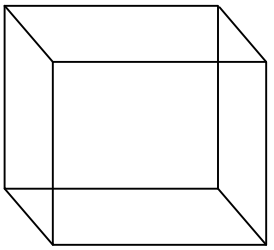


Sec 9.1 Math in 3-d!!

The distance formula in 3d.

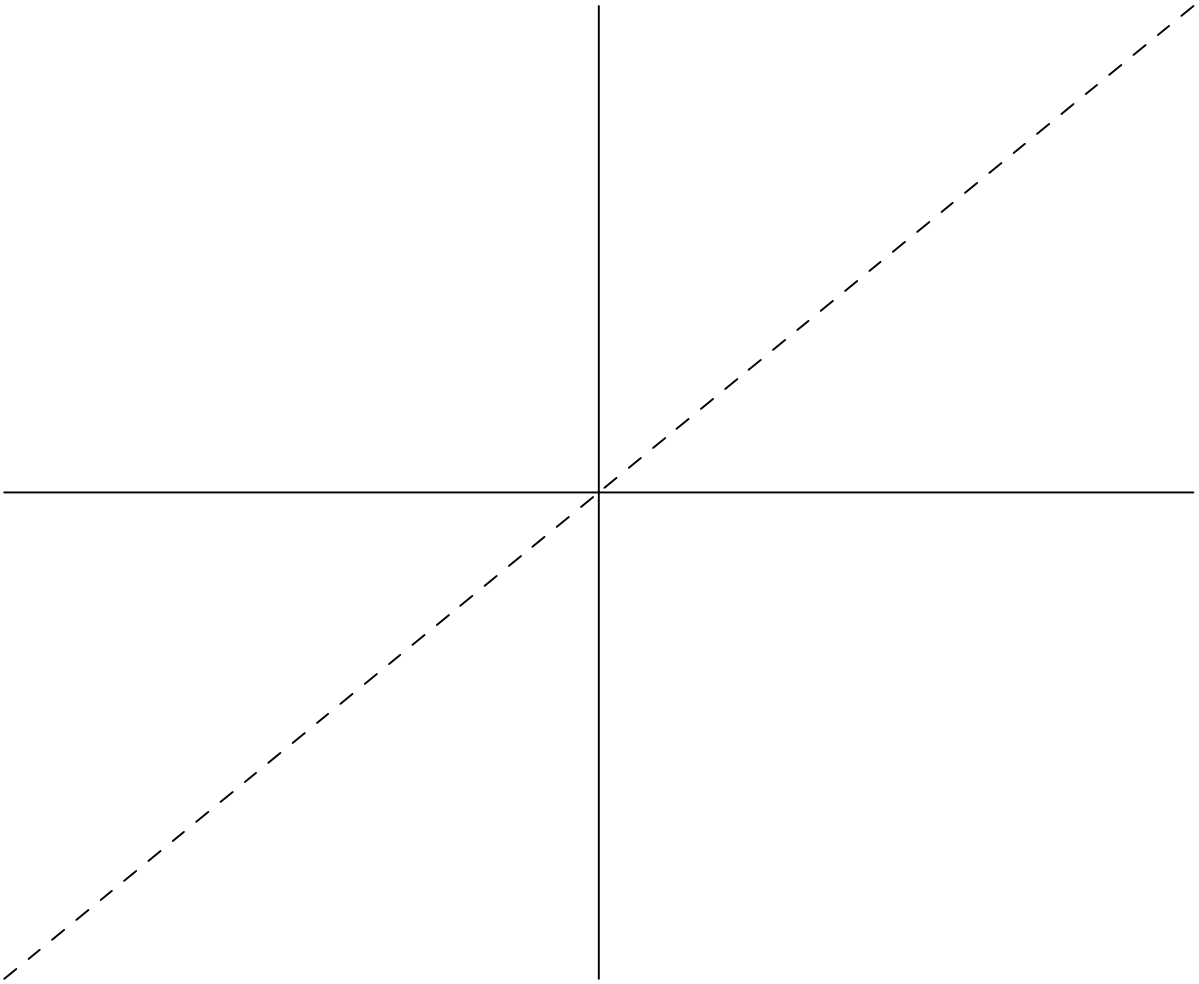






Sec 9.1 Math in 3-d!!

The equation of a sphere in 3d. Just like with a circle we define the equation of a sphere to he the set of all points equidistant from that of the center.



Equation of a Sphere An equation of a sphere with center $C(h, k, l)$ and radius r is

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

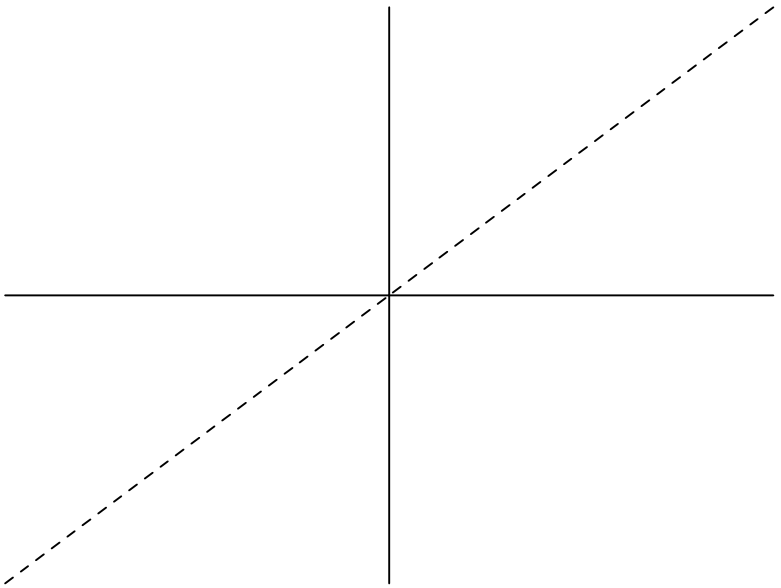
In particular, if the center is the origin O, then an equation of the sphere is

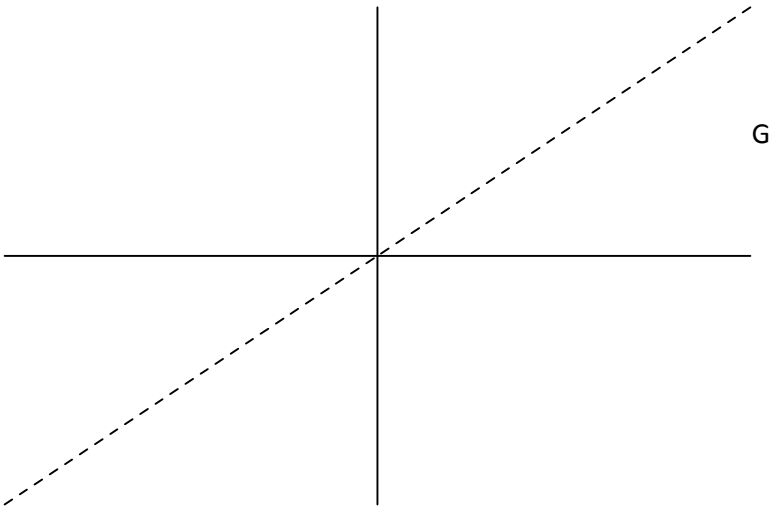
$$x^2 + y^2 + z^2 = r^2$$

Sec 9.1 Math in 3-d!!

Other equations in 3-d

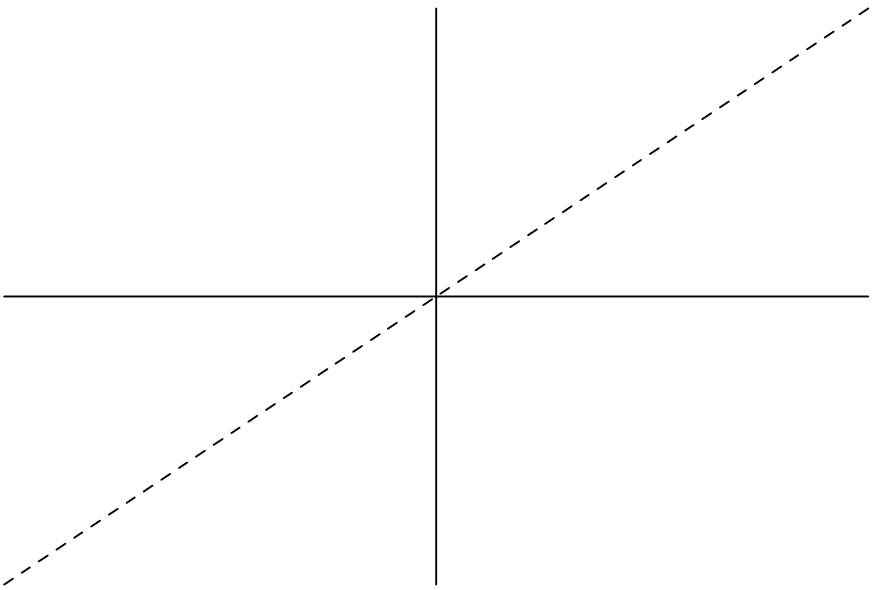
Ex. Graph the equation z=3 in the 3d coordinate system





Graph the equation defined by $x^2 + y^2 = 2$, $z = 2$

Graph the surface defined by $x^2 + y^2 = 4$

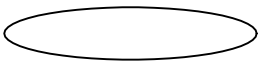


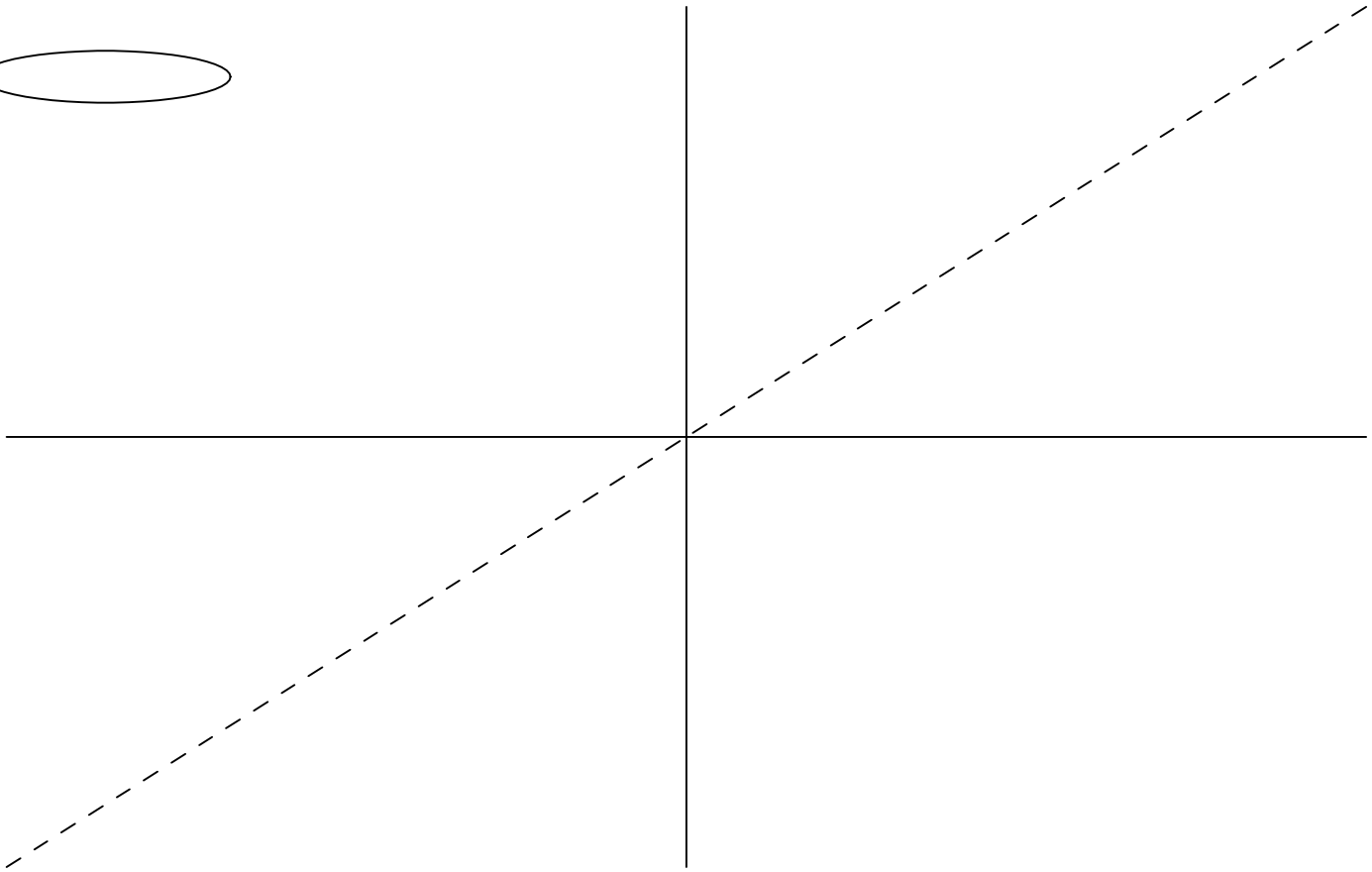
Sec 9.1 Math in 3-d!!

Lets try to graph $z = x^2 + y^2$

This would be easy if z was a fixed number, one way we can accomplish our goal is to pick fixed values for z and see what the graph looks like for each one. We call these level sets……

z	Surface





http://www.livephysics.com/tools/mathematical-tools/online-3-d-function-grapher/?xmin=-4&xmax=4&ymin=-4&ymax=4&zmin=Auto&zmax=Auto&f=x^2%2By^2

Sec 9.1 Math in 3-d!!

Ex Hw #13 Graph the surface $x^2 + y^2 + z^2 - 6x + 4y - 2z = 11$